Sigil

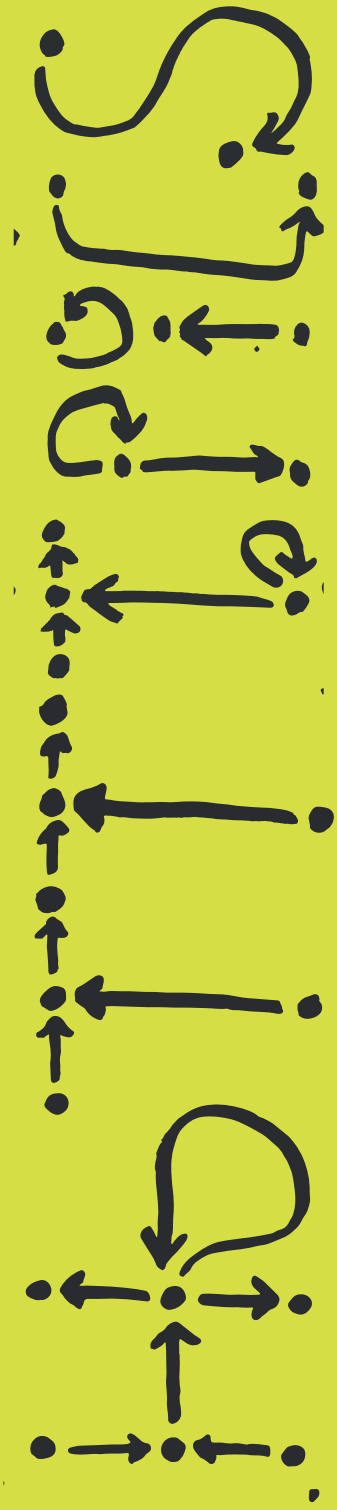
"⟦...⟧ as one gathers one of those magical, closed water lilies, which spring up all of a sudden, enveloping nothingness with their hollow whiteness ⟦...⟧"
–S. Mallarmé

Something is shaking in the so-called "foundations" of mathematics.

One of the most basic mathematical notions, the notion of equality – symbolized by means of the equal sign '=' – is currently undergoing a radical reconceptualization in the framework of a mathematical theory known as homotopy type theory. The mathematical (and political) notions of equality and identity have always taken on a mysterious allure: whereas a proposition a = b stating that two different entities are equal seems to be a contradiction, a proposition a = a stating that an entity is equal to itself – that is, the so-called principle of identity – seems to be nothing but a mere triviality. As we have seen, homotopy type theory allows us to get between contradictory equality and trivial self-identity by revisiting the sharp opposition between the notions of identity and difference from a fresh (topological) perspective. In particular, homotopy type theory shows that self-identity becomes non-trivial only when it envelops a hole, an "extimate" absence, a bounded chunk of nothingness.

A privileged operation by means of which we can introduce equalities is abstraction, that is, the capacity to abstract from certain differences that can be methodically considered irrelevant in a context-dependent manner. By "forgetting" such differences we "equalize" entities that are strictly speaking different. Whereas bracketing differences is a necessary operation in order to afford the otherwise overwhelming complexity of experience, abstraction might also entail forms of violence exerted against all the differences that are abstracted away. If abstract concepts result from an act by means of which we methodically "forget" certain differences, the risk concomitant with abstraction is that of forgetting that we have forgotten these differences.

In this introductory workshop addressed to non-scientists, we explored the possibilities of higher-dimensional thinking, we carnivalistically trained our capacities to stretch strict identities, we gathered some translucent holes in a bamboo forest, and we learned how to cope with the pitfalls of "vicious abstractionism" (W. James). By capitalizing on the topic of this year, we aimed to experiment around "politics of equality" associated with linguistic differences, gender differences, and cognitive diversity. As in the previous editions, we shall explore strategies to deepen the rootedness of science in the material, political, aesthetical, emotional, and spiritual dimensions of the concrete life that both makes science possible and provides the ultimate horizon for its deployment. All in all, we shall not give up on the commitment that orients this overall project: "to serve science only to the extent that science serves life" (F. Nietzsche).



There was a baby born named Ida. Its mother held it with her hands to keep Ida from being born but when the time came Ida came. And as Ida came, with her came her twin, so there she was Ida-Ida.

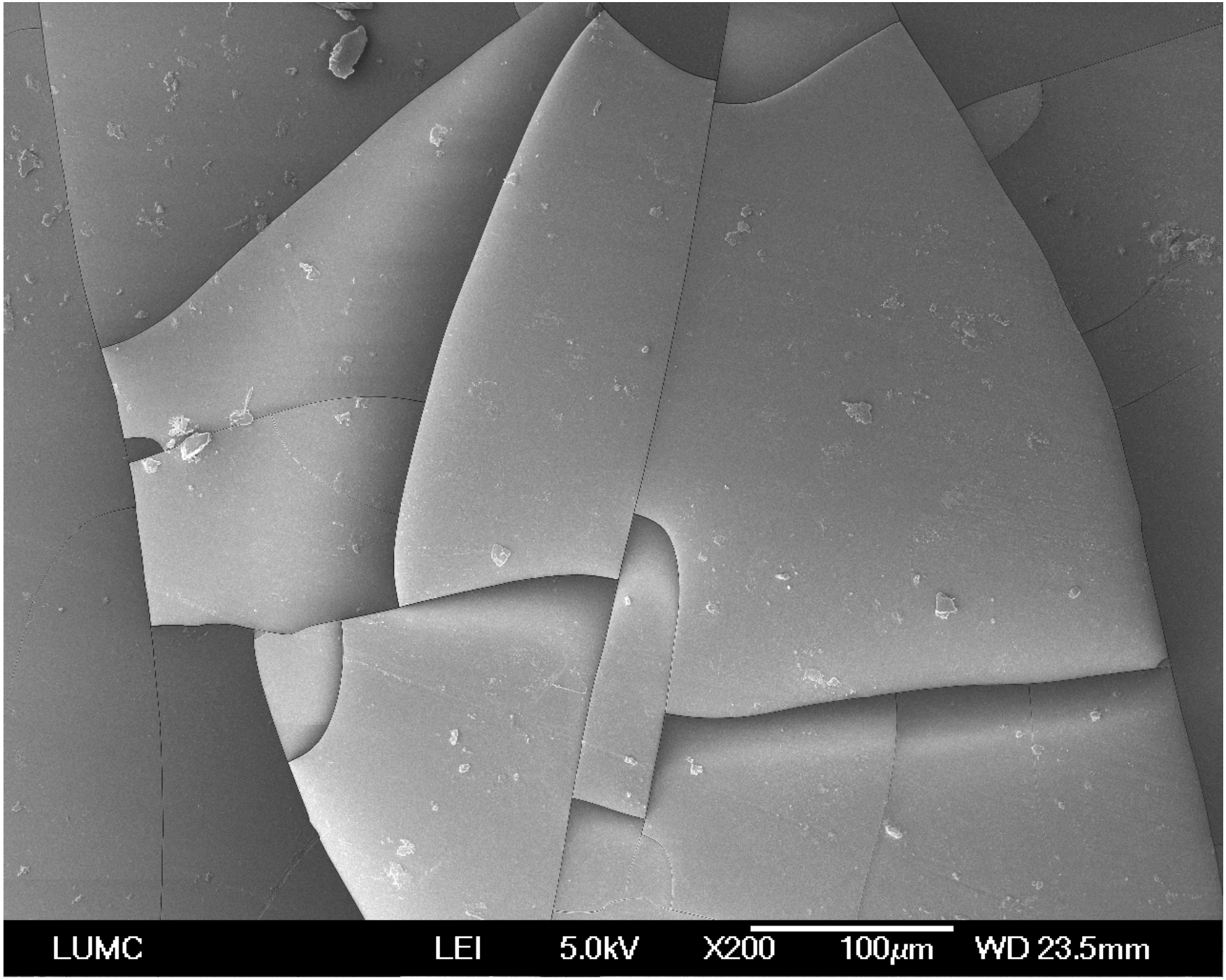
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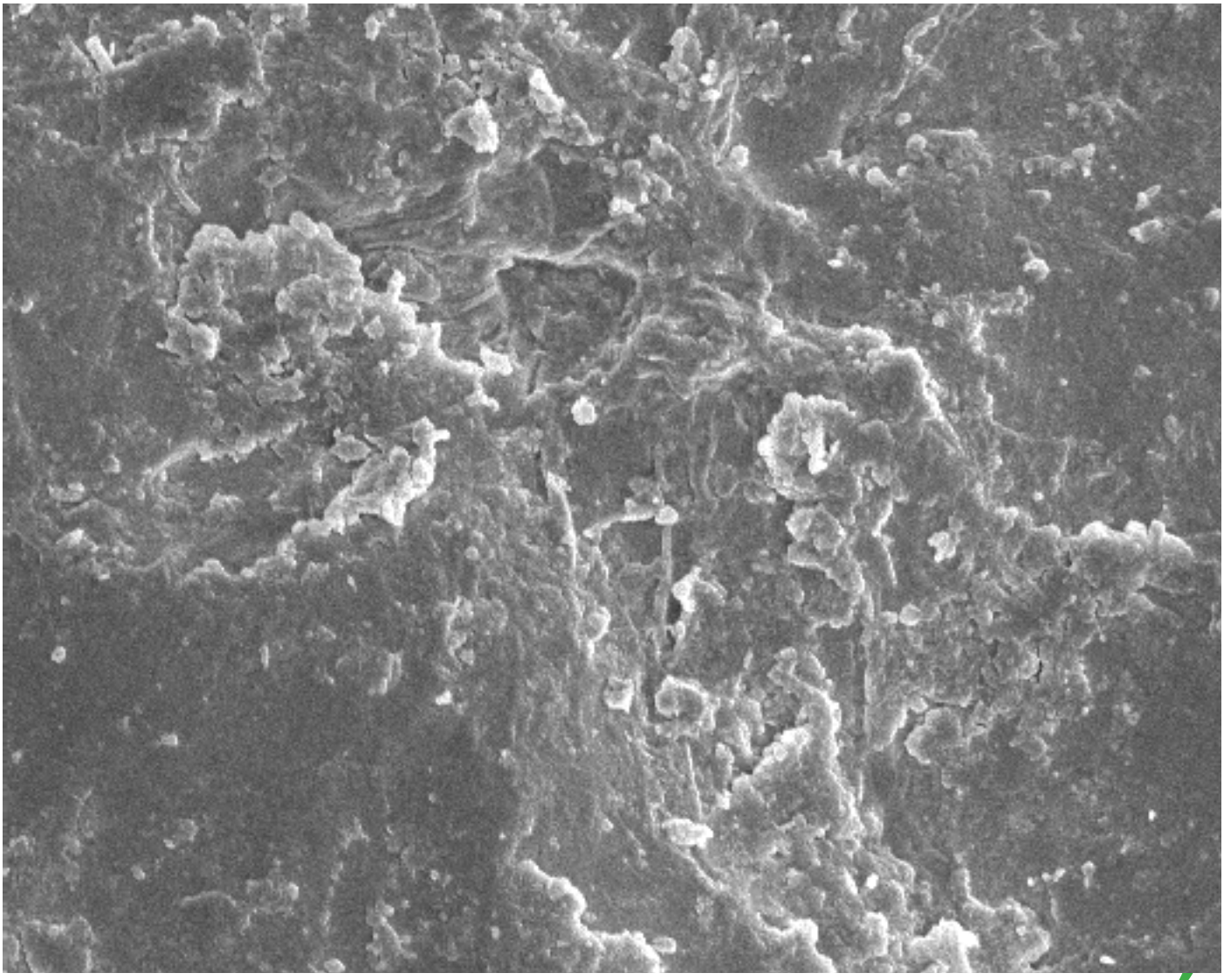
Ida: a novel
Gertrude Stein.



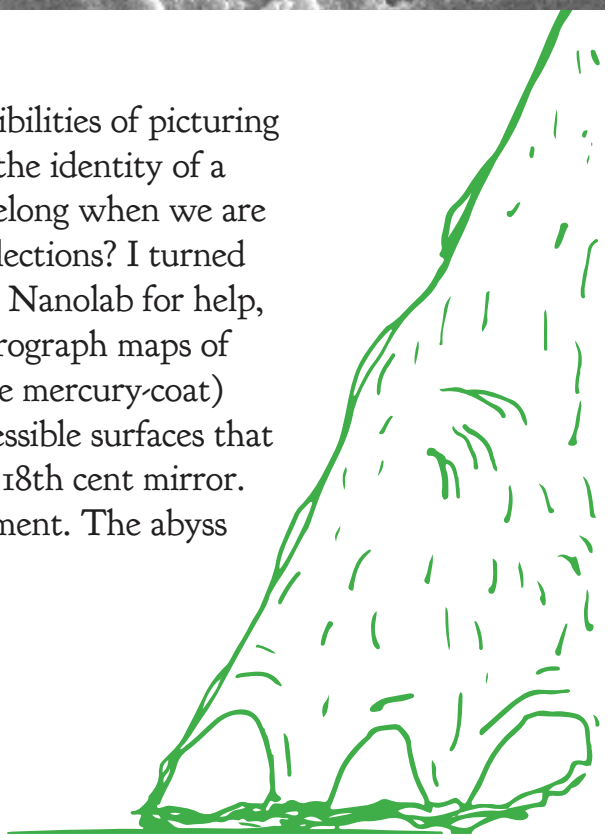





LUMC
LEI
5.0kV
X200
100µm
WD 23.5mm



I wanted to understand the possibilities of picturing a mirror without light. What is the identity of a mirror? To which type does it belong when we are no longer speaking of photon reflections? I turned to the good people of the Leiden Nanolab for help, in order to produce electron micrograph maps of the glass and its back (in this case mercury-coat) surface, the two micrograph-accessible surfaces that constituted this (little chunk of) 18th cent mirror. These are the results. Enlightenment. The abyss stares back. And it is dirty.



SLOGANS OF THE HOTT WORKSHOP

Gabriel Catren

-"Form is emptiness (śūnyatā), emptiness is form." (The Heart Sūtra)

-STRUCTURALIST CREDO:

The relations determine the relata.

-CONSTRUCTIVIST CREDO:

No truth without proof.

-CATEGORY-THEORETICAL ANARCHIST CREDO:

Let's replace transcendent laws with immanent coherators!

-No Entity without Type (P. Martin-Löf).

-No Type without Equality (P. Martin-Löf).

-No Entity without Identity (W.V.O. Quine).

-If there were something in nothingness, then everything would be true.

-The One is so holeless that it cannot even not be.

-Emptiness is spherical.

« Si l'A est tant A que ça, qu'il y reste ! Pourquoi le séparer de lui-même pour si vite le rassembler? », J. Lacan, Séminaire IX, L'identification.

-Without holes, identity is trivial.

-« [[...]] ces cercles qui font le tour du trou du tore », J. Lacan, Séminaire IX, L'identification.

-« [[...]] en el vacío se puede esconder un canguro sin perder su saltante júbilo. », J. Lezama Lima.

-An entity singularly exhibits the emptiness it partially fills.

-There is no equality without common property.



GRASPING PROOFS OF LOVE

MAARTEN VAN DEN BUSSCHE

a rose in this garden
has its stem snapped
unbroken but bent into
a state of ending with
juices still flowing yet
equally already dead

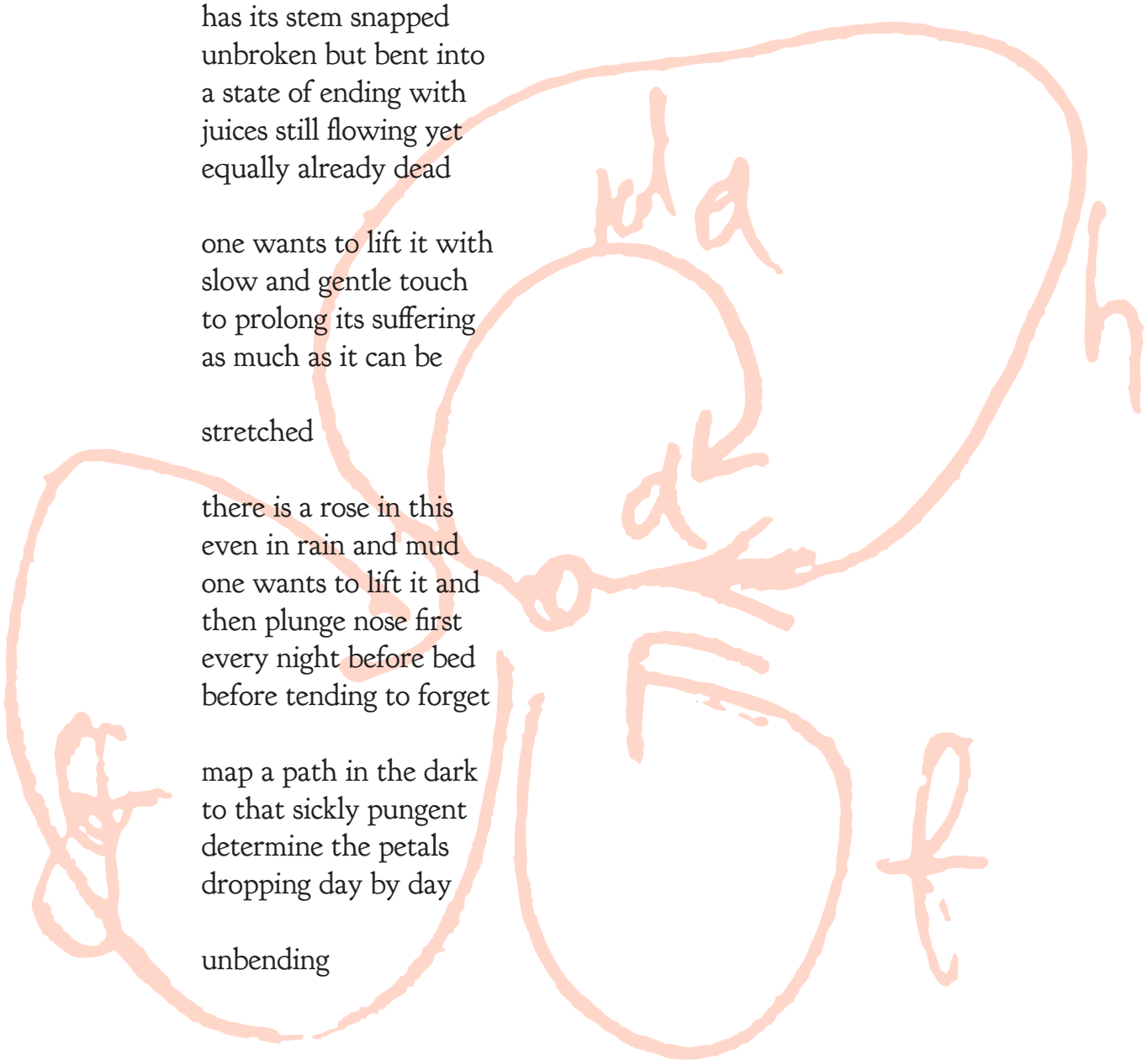
one wants to lift it with
slow and gentle touch
to prolong its suffering
as much as it can be

stretched

there is a rose in this
even in rain and mud
one wants to lift it and
then plunge nose first
every night before bed
before tending to forget

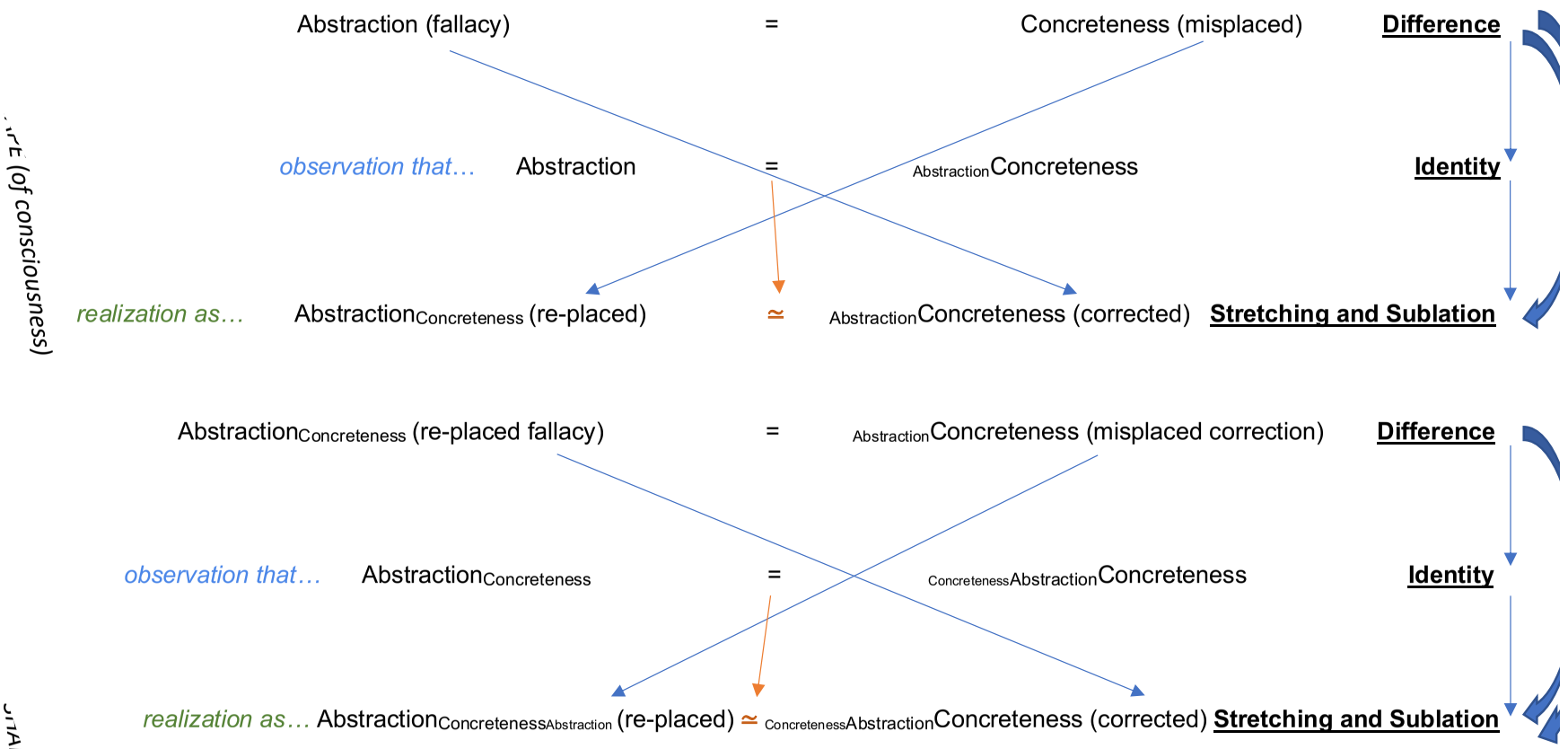
map a path in the dark
to that sickly pungent
determine the petals
dropping day by day

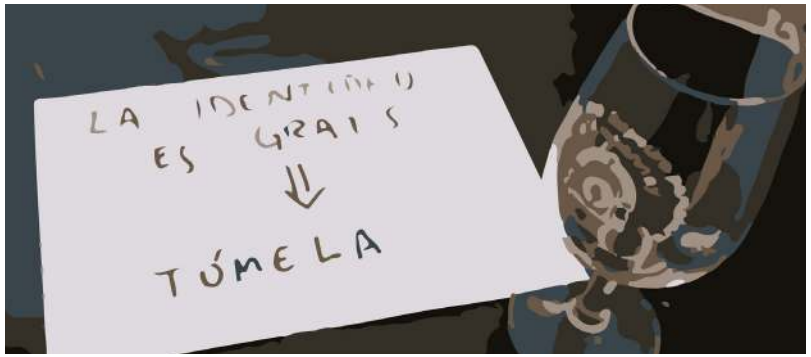
unbending



"THERE'S NO PAF BETWEEN A AND B
IF THEY RESIDE IN THE SAME PLACE."

A diagrammatic sketch at the dialectic of abstraction and (misplaced) concreteness, in the spirit of HoTT





GROUPS

Group : Set, function, neutral element
$(X, *, e)$ structure

$a * e = a$

$a * a^{-1} = e$

Set, tous les objets avec lesquels on peut travailler dans une catégorie d'opérations

Structured dust

$\mathbb{Z} = \{(..., -2, -1, 0, 1, 2...), +, 0\}$

$2 + 0 = 2$

$2 + (-2) = 0$

A set has no geometrical/spatial value (Ø notion of distance)

Topological space: $(X, \mathcal{T})$

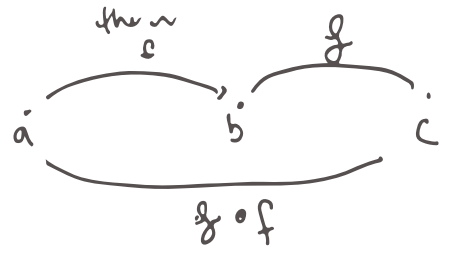
specification of unbreakable part; neighborhood

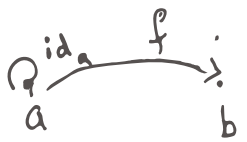
discrete vs indiscrete topology
↓ ↓
no community no intimacy.

Cat. theory : the relations determines the relata.
↳ Structuralist credo

Category : set of objects with "morphisms" or relations between

Composition





fulfills associativity

$f \circ id_a = f$

Manière d'assembler des groupes par composition -> catégorie

Morphisms are category dependant.

object : bundle of properties

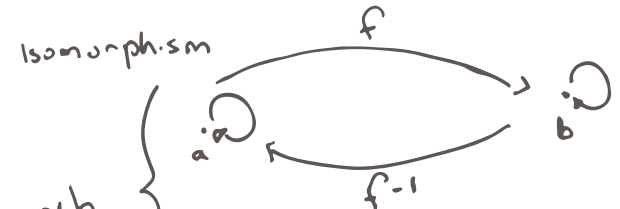
Principle of indiscernability of equals

to 2 equal objects should have same properties

properties invariant across equalities

[invertible functions]

isomorphism



-> composable

↓

category dependant

$a \simeq b$ { $f \circ f^{-1} = id_a$ $\quad f \circ f^{-1} = id_b$

going outside to find identity to oneself) or $a = a$ staying in definition by environment.

- Sameness is category dependant

from strict equalities to isomorphisms

$a = b \longrightarrow a \simeq b$

properties have to be invariant under isomorphism.

have same relation with the rest of the category

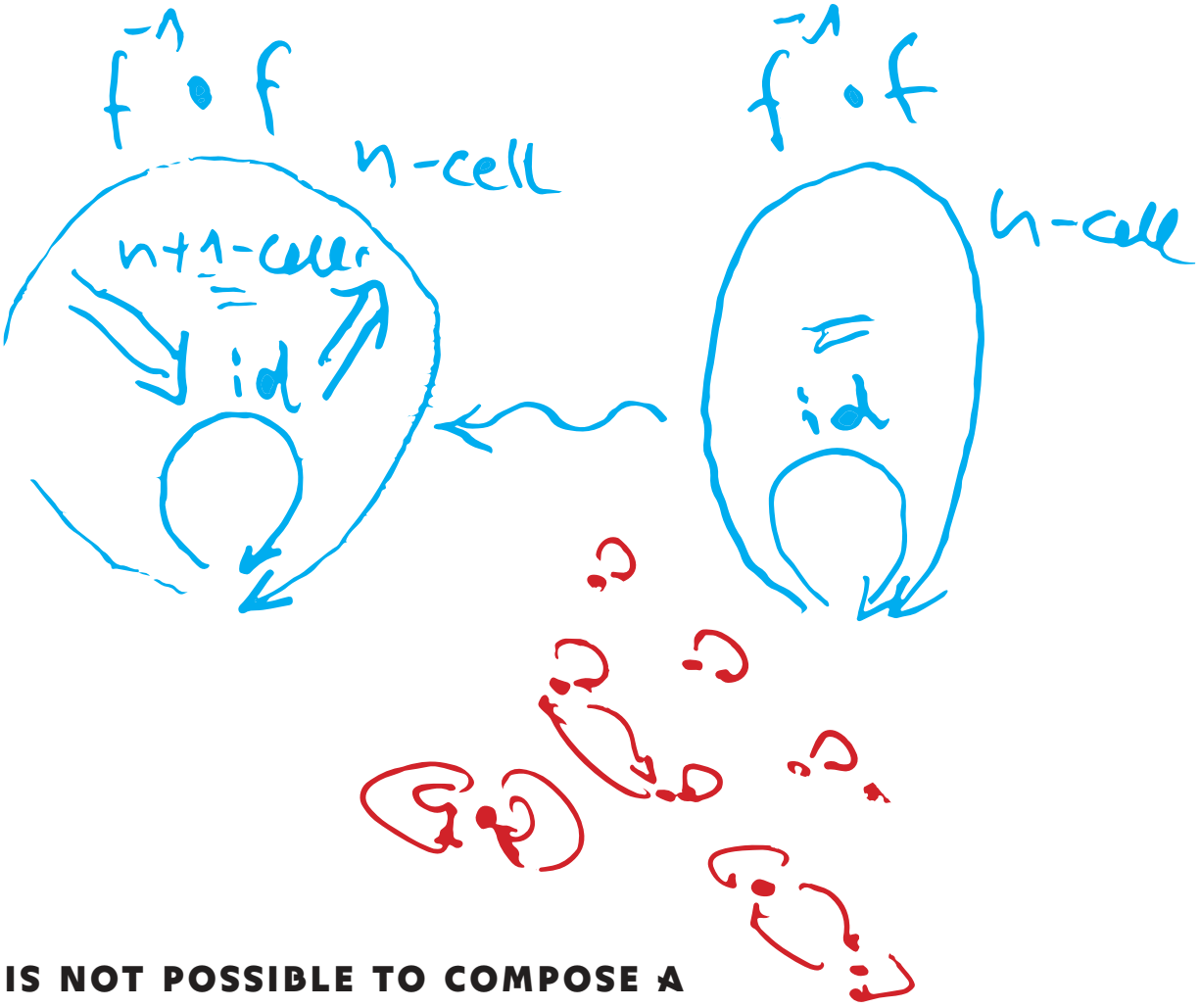
object that rotates -> same but you can go back to

Relating morphisms nécessite strict equalities

Morphism, composition, functions, continuous functions

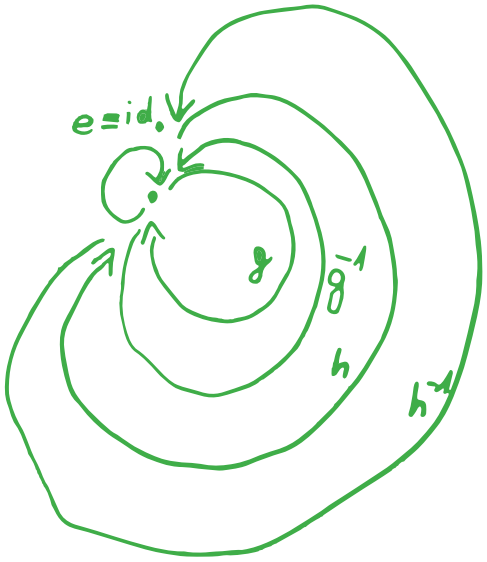
↓ ↓ ↓ morph in ↓

2x morphism conditions of assemblage — morph in sets — to topological space

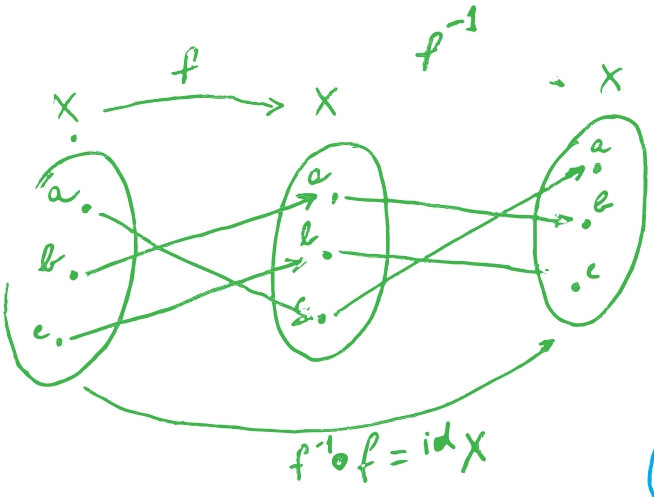


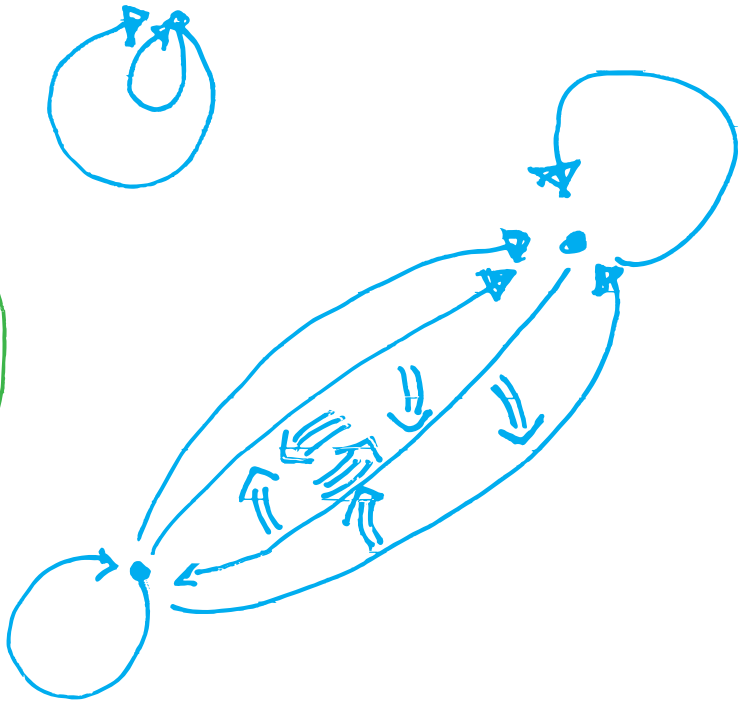
"IT IS NOT POSSIBLE TO COMPOSE A PAF BETWEEN THEM BECAUSE THERE'S A HOLE BETWEEN THEM."





Objects from a group become morphisms in a group as a category









SETS

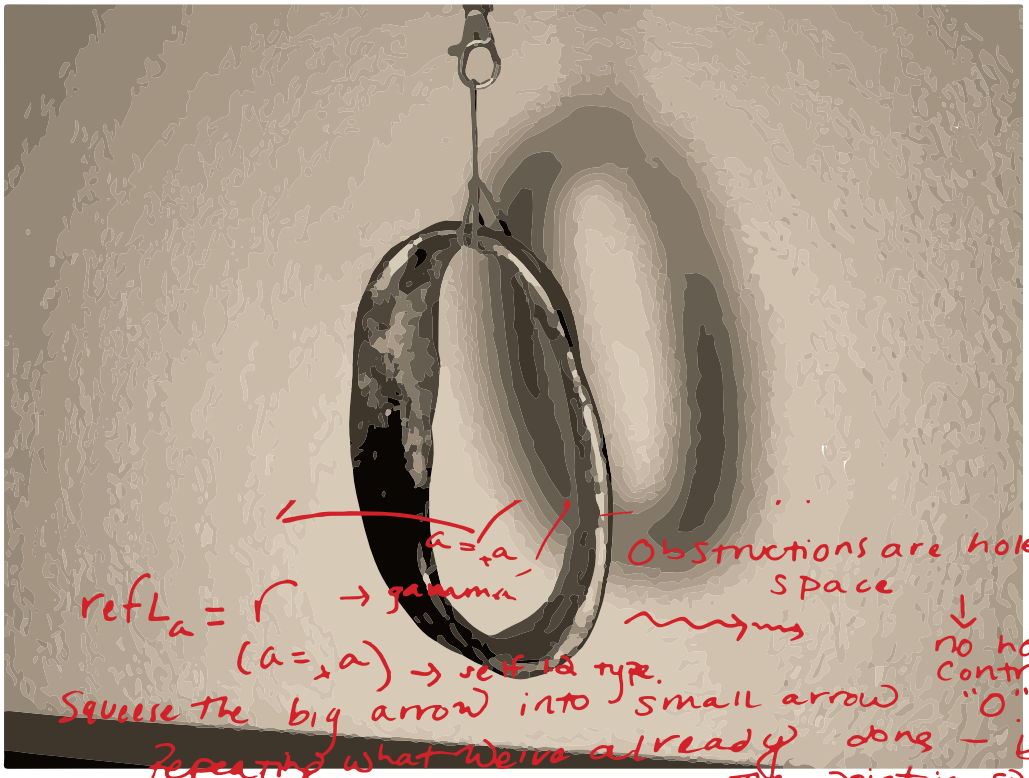
∞ - GROUPOIDS

"="

"SPACES"

(HOMOTOPY TYPES)

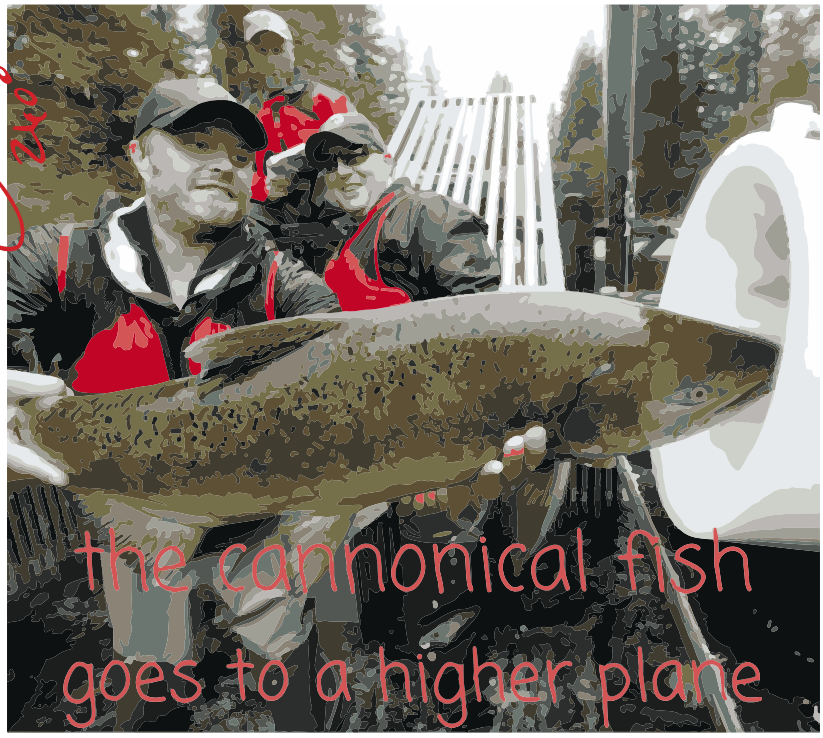








"THEY ARE PAF CONNECTED."


the cannonical fish
goes to a higher plane


Daniel Radcliffe Discusses Comparisons To Elijah Wood: "Conceptually, We Are The Same"
GameSpot 2d
"Mathematics may be defined as the subject
in which we never know what we are talking about."
Bertrand Russell


iso-morphing power rangers

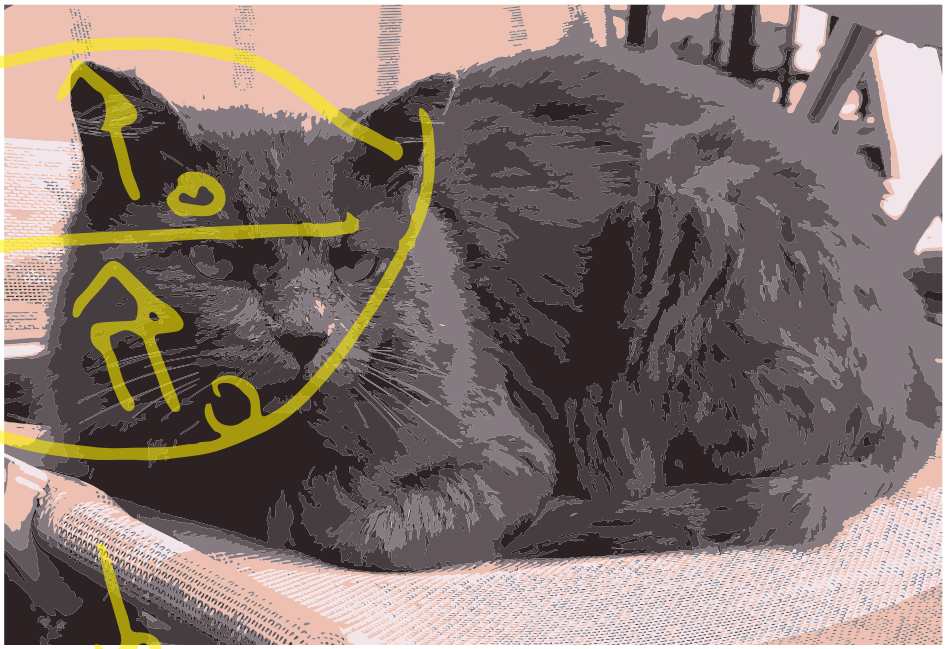
BEHIND THE CAT

there is a cat
and behind this cat
another cat
waiting to what

in the middle of the ocean
there is a boat
and under this boat
surrounded by the blue
there is a person

none of the cats
moves one single muscle
but cat c starts thinking
if behind him is there another cat

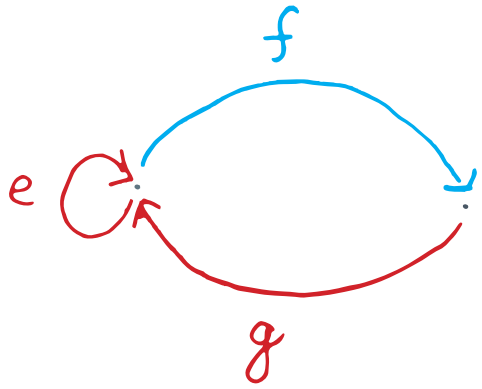
for the person
under the boat
in the middle of the ocean
the boat is one shadow



MY NAP IS OVER/LIFE IN THE CHAPEL ROOM (A TANKA FOR DADO AND VIVIANA)

The sound of voices
singing togetherness through
reason; feet on the
floor pitter patter and play;
soft hands guide me to waking.

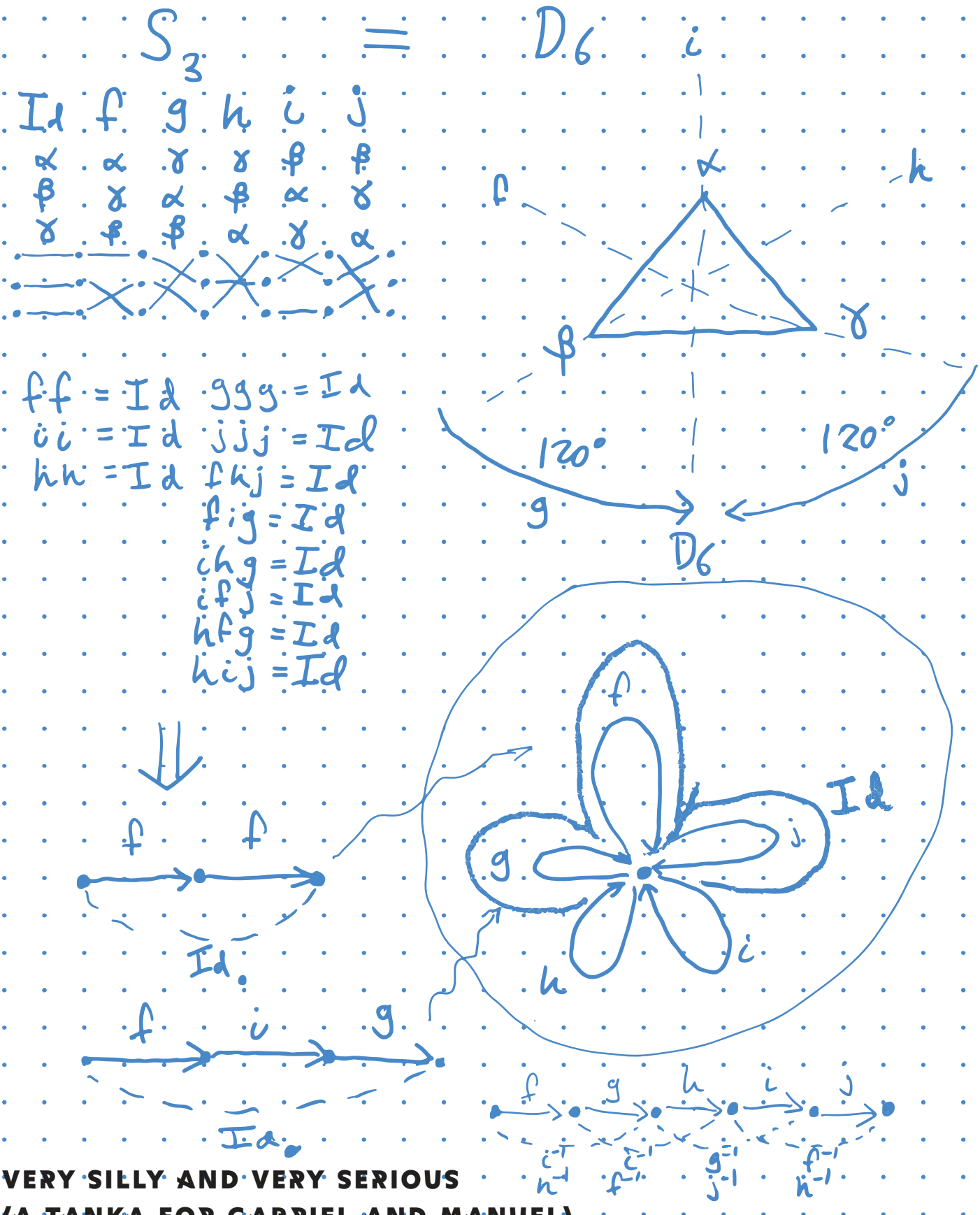
any morphism f within a category for which another morphism g exist whose source matches the target of the former and whose target matches its source, implies the existance of another arrow e from the source of the first arrow to the source again*. This last arrow is called an **endomorphism**.



however, this **does not** imply that f is **isomorphic**. that is, g is not necessarily the inverse of f. therefore e is not necessarily an **identity** morphism.

* which is the composition of g and f.

"HERE AM SPEAKING ABOUT PAF, WHICH HAS AN INTUITIVE MEANING."



VERY SILLY AND VERY SERIOUS (A TANKA FOR GABRIEL AND MANUEL)

I wake up twenty
minutes late for the first day
of class, which means I
have resigned my place; this is
how I have always studied.

SER O ESTAR,

ésa es la pregunta.

Ser
una cosa, estar
otra.

Siendo sólo eso, está difícil, estático.

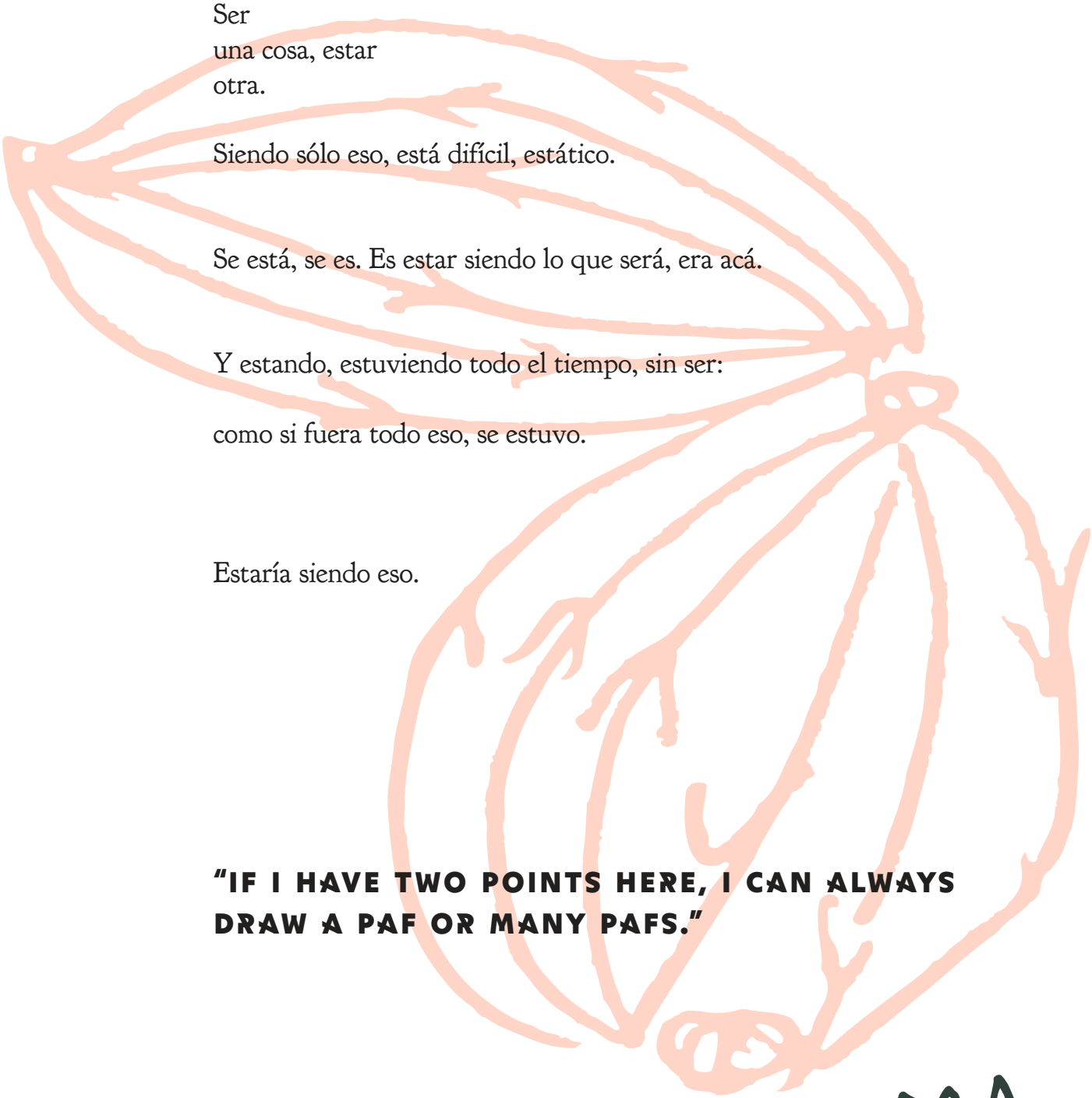
Se está, se es. Es estar siendo lo que será, era acá.

Y estando, estuviendo todo el tiempo, sin ser:

como si fuera todo eso, se estuvo.

Estaría siendo eso.

"IF I HAVE TWO POINTS HERE, I CAN ALWAYS DRAW A PAF OR MANY PAFS."



MAGMA

WHAT I LEARNED WHILE SHARING A BUILDING AND A DAILY DINNER WITH 40 PEOPLE BEING ENGAGED IN MATHEMATICS AND PHILOSOPHY FOR ONE WEEK:/ WAT IK HEB GELEERD TERWIJL IK EEN WEEK LANG EEN GEBOUW EN EEN DAGELIJKS DINER DEELDE MET VEERTIG MENSEN DIE ZICH BEZIGHIELDEN MET WISKUNDE EN FILOSOFIE:

- Propositions used to be true or false, just like equations. What they're now, I'm not sure.
- Oooooh ooh oh oh oh waaaiejaaaaajo. Aba aba aba.
- The Black Eyed Peas sampled a Brazilian song.
- It takes about 12 hours to travel from Berlin to Paf.
- A deterministic worldview doesn't preclude the idea that nothing is predictable (something about a ball bouncing).
- 'Mind asymmetries' could perhaps be replaced by - 'Assume intelligence', but we're not sure.
- If you speak nothing but European languages, it doesn't necessarily mean you're happy about that.
- I shouldn't say too loudly that I hate astrology.
- The game of life is a program that uses rules to change grids with black and white squares. You can also use it writing Hebrew poetry.
- I don't like to dance in the middle of a circle.
- It's better not to talk about Badiou if you don't know that much about Badiou.
- If you know a lot about Badiou, you better say so whenever someone starts talking about Badiou. But how do you do that without coming across as self-righteous?
- My brain is capable of completely forgetting about a promise I made. I felt anxious after finding out.
- My roommate at home who researches the behavior of numbers might be very smart, but we don't know how to find out.
- Some object's inside is their outside and vice versa.
- Vargas Llosa's literature's supposed to be boring.

Eva van den Boogaard
PAF, 26/10/2023

LOCATION: STELLA MARIS,

since 1950 a non-denominational facility and hospice for the care of psychiatric medical patients in Black River Falls, Wisconsin.

Date: October 27, 1972.

Characters:
Maria, 20-year-old patient of the facility.
Dr. Cohen, 43-year-old psychiatrist of the facility.

Dr. Cohen: You're skeptical about mathematics?
Maria: Yes.
Why is that?
It's a long story.
Are you still working at mathematics?
No. Not what you'd call mathematics.
What sort of mathematics were you doing?
Topology. Topos theory.
But you're not doing that anymore.
No. I got distracted.
What was it that distracted you?
Topology. Topos theory.
Maybe we should skip the mathematics for the time being.
That's fine. I didn't know what I was doing anyway.
I'm surprised to hear you say that. Couldn't you get help from some of the other mathematicians?
No. They didn't know either. When you get to topos theory you are at the edge of another universe. You have found a place to stand where you can look back at the world from nowhere. It's not just some gestalt. It's fundamental.
What about your friend?
Grothendieck.
Yes.
Well. The topos theory he came up with is a witches' brew of topology and algebra and mathematical logic. It doesn't even have a clear identity. The power of the theory is still speculative. But it's there. You have a sense that it is waiting quietly with answers to questions that nobody has asked yet.
That sounds a bit platonic.
Don't it though?
So what about him?
He doesn't write me anymore.
Why is that?

Mostly because he's given up mathematics.

As have you.

Yes. As have I.

Was that hard?

Well. I think maybe it's harder to lose just one thing than to lose everything.

One thing could be everything.
Yes. It could.
Are you crying?
I'm sorry.
But he gave it all up.
Yes.
Why?
His friends believe that he has become increasingly unstable mentally.
Has he?
It's complicated. You end up talking about belief. About the nature of reality.

(Excerpt from Stella Maris, Cormac McCarthy.)

"A TERM WITH THIS EQUALITY TERM IS A PAF."



"Holotropy joke"

THE PROJECT OF IDENTITY

What is the meaning of the arrow Id in the categorical (monoidal) representation of the group S_3/D_6. As an uncentered group, the very notion of a locus around which the composition of group elements would circulate is lacking a concept. The pure · stands for something like the mere syntactical agreement of composition. Further, the Id written down multiple times in the diagram provided is only something like the highest common factor of all the group elements. As such it doesn't exist, a shorthand for a coordination of the identity of differences. Taking certain routes of a journey back to where it first found itself, the origin can take a blind alley.

S_3/D_6 as the permutations of a (3,0) category (read "with ~Id~"□), or the symmetries of an equilateral triangle under reflexion and rotation, only happen to satisfy the group axioms insofar as the inverses (sometimes non-trivial!), are satisfied by operations immanent to the domain in question. The logical requirement of inverses has been invented to categorise this group, not define it in any simple sense.

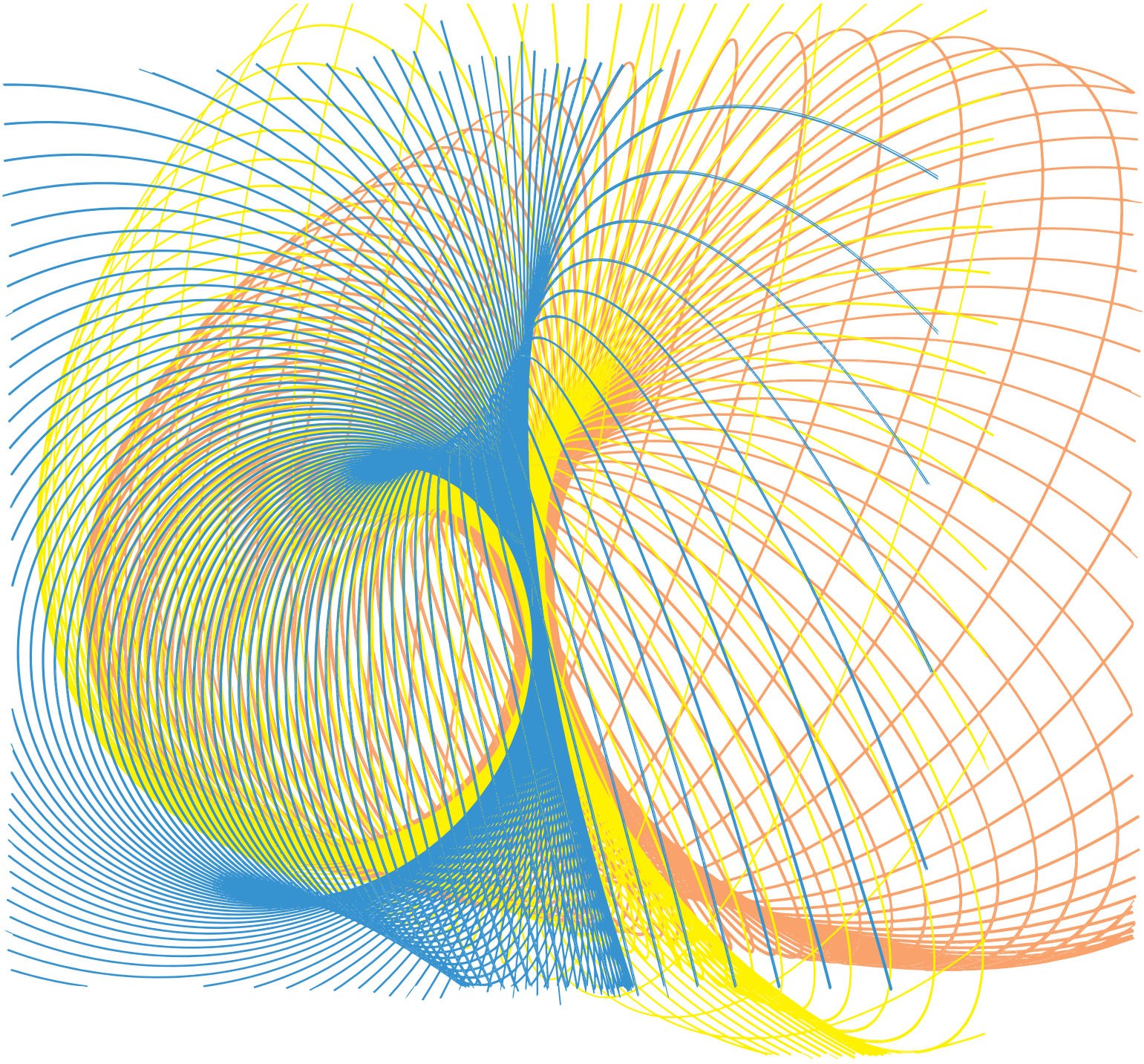
While the totally disextensional nature of category theory precipitates into an absolutist intensionality, contemporary philosophy has already recognised the dialecticisation of notions within this domain. What of the semi-metric spaces of identity of objects intensional worlds (see Logics of Worlds) whose operations, furthermore, take place in a real-valued logic and not a intuitionism formalised in a classical world. What of pointless spaces, paraconsistent logics, alternative axiomatics of set theory; what is the relaxation of equality to an equivalence in constructive space compared to the relaxation of an axiom in classical space?

Isn't the identity arrow itself a residuum of set theory in the categorical tradition, whereby each element (via extensionality) is equal to itself. Paths that fade—what is "'this' piece of paper on which I am writing" (PdG para 24).

Edward Wilson

GRANDMASTER FLASH(ES) OF THE PAF KITCHEN (A TANKA FOR CHRISTIAN AND MATT)

Christian and Matt
dance around the kitchen and
each other, sharing
their knowledge of nourishment
as if they each knew nothing.



HOPF FIBRATION

∀
∀
∀

LIFE IS NOT A STABLE CATEGORY

it must be constructed, performed

an infinite space
of places, paths
shapes, identities
loops, lovers, laws

i am tired
doing nothing is doing something

there is music playing in the kitchen
some people are talking in the hallway
another is on a sofa, reading

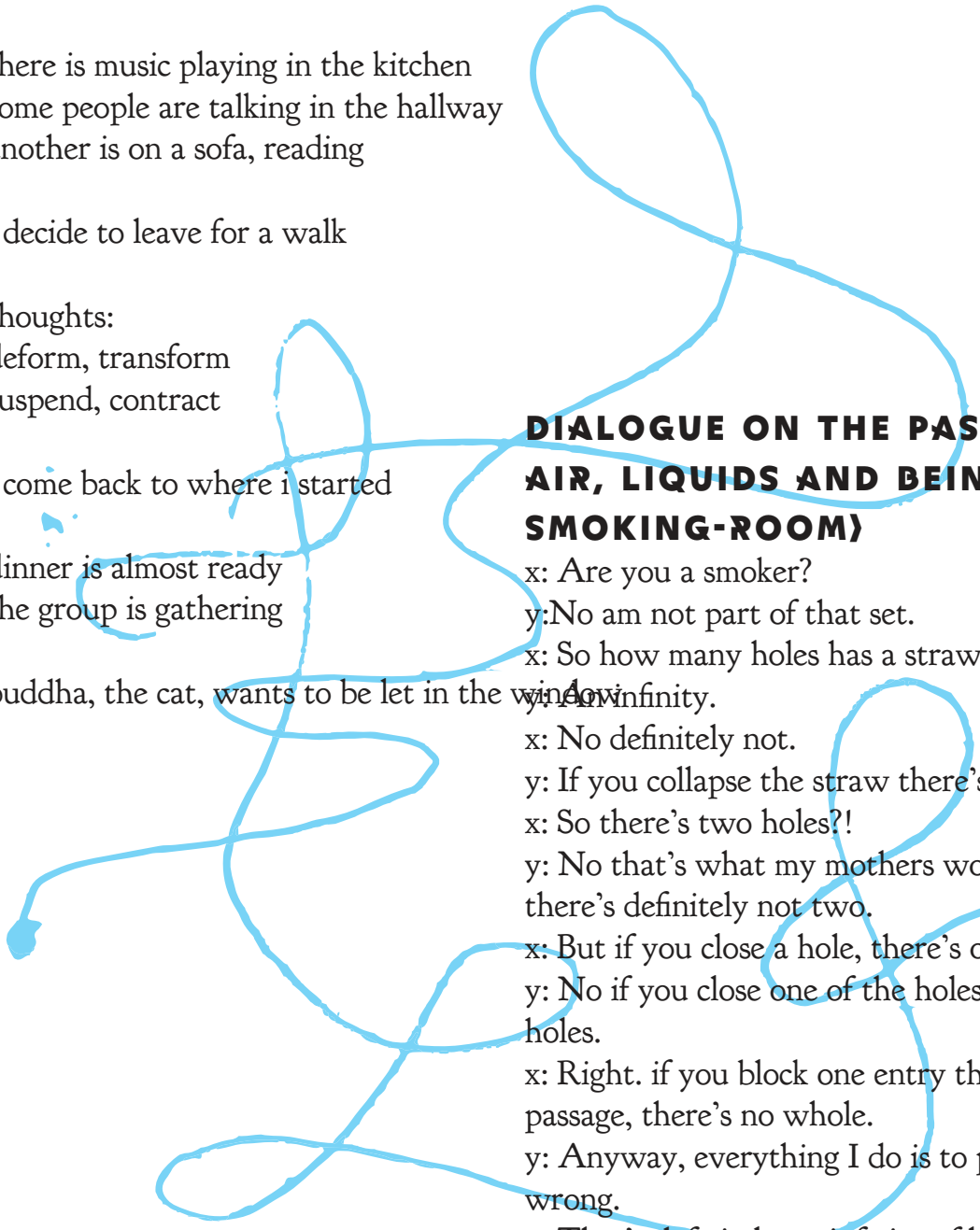
i decide to leave for a walk

thoughts:
deform, transform
suspend, contract

i come back to where i started

dinner is almost ready
the group is gathering

buddha, the cat, wants to be let in the window



DIALOGUE ON THE PASSAGE OF AIR, LIQUIDS AND BEINGS (IN THE SMOKING-ROOM)

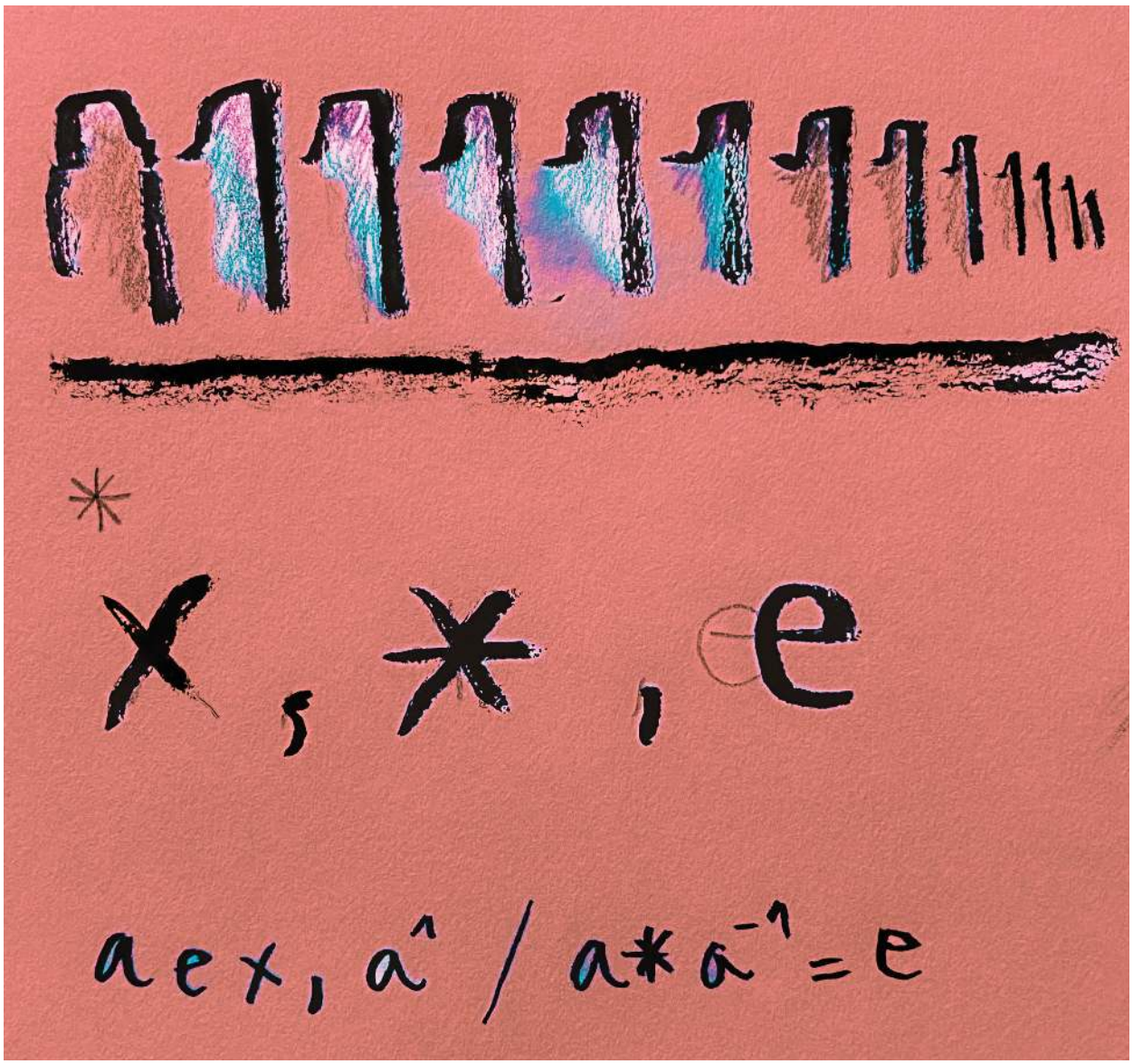
x: Are you a smoker?
y:No am not part of that set.
x: So how many holes has a straw? one or two?
y: An infinity.
x: No definitely not.
y: If you collapse the straw there's only one hole.
x: So there's two holes?!
y: No that's what my mothers would say, so there's definitely not two.
x: But if you close a hole, there's only one left.
y: No if you close one of the holes, there's no more holes.
x: Right. if you block one entry there's no more passage, there's no whole.
y: Anyway, everything I do is to prove my mum wrong.
x: That's definitely an infinity of holes. Have a sip.

Esto es un verso

Esto es un verso.

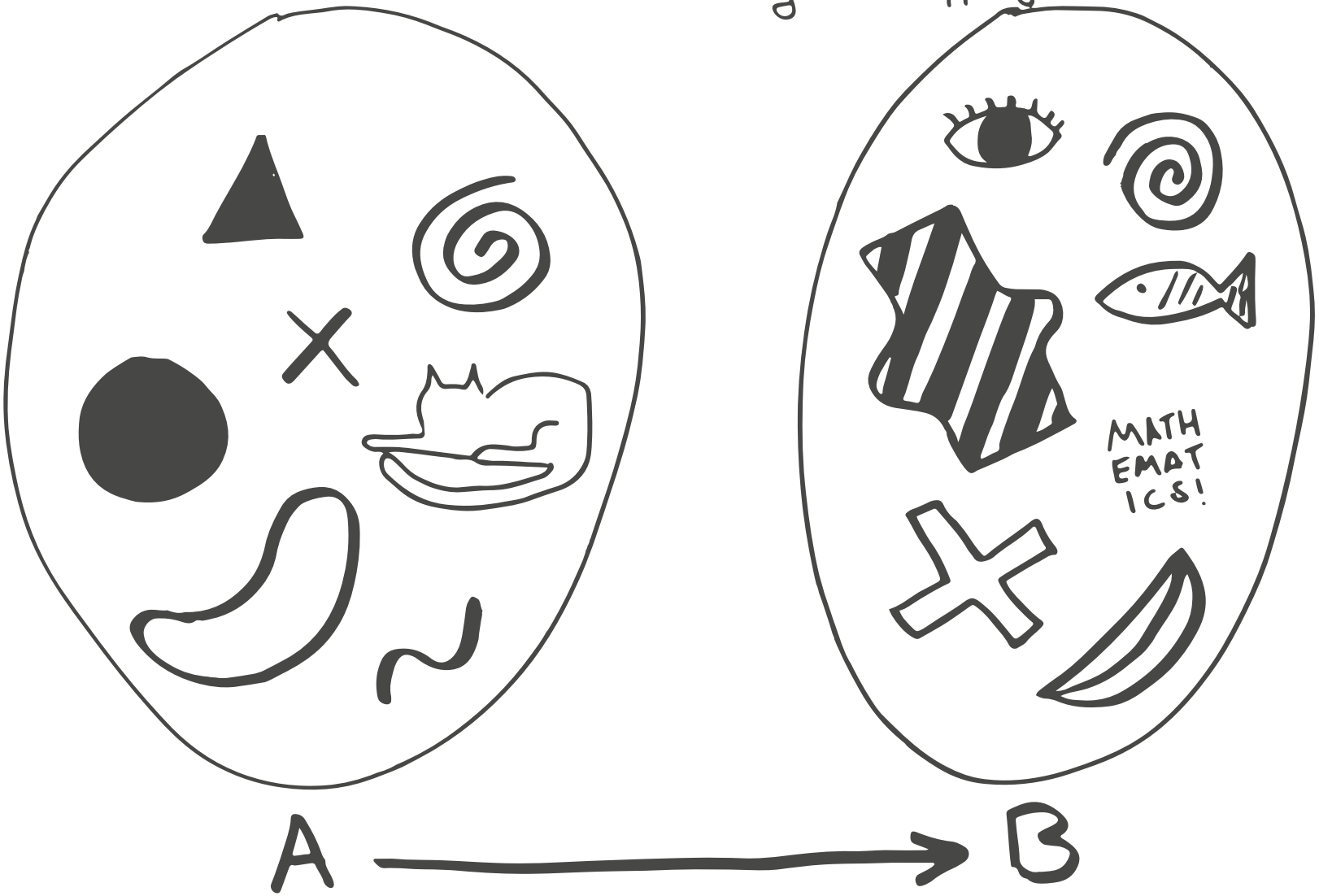


Mentira.



Map Your Future

1. Map objects in A to B by drawing arrows
2. Turn the page and interpret your mapping



INDEX OF CONTRIBUTORS



If she said anything she said yes. More than once nothing was said. She said something. If nothing is said then Ida does not say yes. If she goes out she comes in. If she does not go away she is there and she does not go away. She dresses, well perhaps in black why not, and a hat, why not, and another hat, why not, and another dress, why not, so much why not.She dresses in another hat and she dresses in another dress and Andrew is in, and they go in and that is where they are. They are there. Thank them.Yes.

Ida: a novel
Gertrude Stein

PAF (PERFORMING ARTS FORUM)
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